

```

*****
*
*      STAAD.Pro
*      Version 2003      Bld 1003.ASIA
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date=      MAR 3, 2012
*      Time=      18: 2:25
*
*      USER ID: NCK Associates
*****

```

INPUT FILE: Substation Frame - R2.STD

```

1. STAAD SPACE
2. START JOB INFORMATION
3. ENGINEER DATE 03-MAR-12
4. END JOB INFORMATION
5. INPUT WIDTH 79
6. UNIT METER KN
7. JOINT COORDINATES
8. 1 0 0 0; 2 2.4 0 0; 3 0 0 1.5; 4 2.4 0 1.5; 5 0 2 0; 6 2.4 2 0; 7 0 2 1.5
9. 8 2.4 2 1.5; 9 0 0 3; 10 2.4 0 3; 11 0 2 3; 12 2.4 2 3; 13 0 0 4.5
10. 14 2.4 0 4.5; 15 0 2 4.5; 16 2.4 2 4.5; 17 0 0 6; 18 2.4 0 6; 19 0 2 6
11. 20 2.4 2 6; 21 3.9 0 0; 22 3.9 0 1.5; 23 3.9 2 0; 24 3.9 2 1.5; 25 3.9 0 2.1
12. 26 3.9 2 2.1; 27 3.9 0 3; 28 3.9 2 3; 29 3.9 0 4.05; 30 3.9 2 4.05; 31 3.9 0 6
13. 32 3.9 2 6; 33 2.4 2 4.05; 34 6.2 0 0; 35 6.2 2 0; 36 6.2 0 1.5; 37 6.2 2 1.5
14. 38 6.2 0 2.1; 39 6.2 2 2.1; 40 8.6 0 0; 41 8.6 2 0; 42 8.6 0 1.5; 43 8.6 2 1.5
15. 44 8.6 0 2.1; 45 8.6 2 2.1; 46 11 0 0; 47 11 2 0; 48 11 0 1.5; 49 11 2 1.5
16. 50 11 0 2.1; 51 11 2 2.1; 52 11 2 3; 53 6.2 0 3; 54 6.2 2 3; 55 8.6 0 3
17. 56 8.6 2 3; 57 11 0 3; 58 6.2 0 4.05; 59 6.2 2 4.05; 60 8.6 0 4.05
18. 61 8.6 2 4.05; 62 11 0 4.05; 63 11 2 4.05; 64 6.2 0 6; 65 6.2 2 6; 66 8.6 0 6
19. 67 8.6 2 6; 68 11 0 6; 69 11 2 6; 70 2.4 0 7.26; 71 2.4 2 7.26; 72 3.9 0 7.26
20. 73 3.9 2 7.26; 74 6.2 0 7.26; 75 6.2 2 7.26; 76 8.6 0 7.26; 77 8.6 2 7.26
21. 79 3.9 5.1 0; 80 6.2 5.1 0; 81 8.6 5.1 0; 82 11 5.1 0; 84 3.9 3.6 0
22. 85 6.2 3.6 0; 86 8.6 3.6 0; 87 11 3.6 0; 89 3.9 5.1 6; 90 6.2 5.1 6
23. 91 8.6 5.1 6; 92 11 5.1 6; 93 3.9 3.6 6; 94 6.2 3.6 6; 95 8.6 3.6 6
24. 96 11 3.6 6; 98 11 5.1 1.5; 99 11 3.6 1.5; 100 11 5.1 3; 101 11 3.6 3
25. 102 11 5.1 4.05; 103 11 3.6 4.05; 104 3.9 5.1 1.5; 105 3.9 3.6 1.5
26. 106 3.9 5.1 3; 107 3.9 3.6 3; 108 3.9 5.1 4.05; 109 3.9 3.6 4.05; 110 11 6.1 6
27. 111 11 5.6 3; 112 11 6.35 7.5; 114 3.9 6.1 6; 115 3.9 5.6 3; 116 3.9 6.35 7.5
28. 117 8.6 6.1 6; 118 8.6 5.6 3; 119 8.6 6.35 7.5; 120 3.8 5.1 0; 121 6.2 6.1 6
29. 122 6.2 5.6 3; 123 6.2 6.35 7.5
30. MEMBER INCIDENCES
31. 1 5 6; 2 7 8; 3 5 7; 4 6 8; 5 1 5; 6 3 7; 7 2 6; 8 4 8; 9 11 12; 10 7 11
32. 11 8 12; 12 9 11; 13 10 12; 14 15 16; 15 11 15; 16 12 33; 17 13 15; 18 14 16
33. 19 19 20; 20 15 19; 21 16 20; 22 17 19; 23 18 20; 24 23 24; 25 21 23; 26 22 24
34. 27 6 23; 28 8 24; 29 25 26; 30 27 28; 31 12 28; 32 24 26; 33 26 28; 34 29 30
35. 35 28 30; 36 31 32; 37 30 32; 38 20 32; 39 33 16; 40 33 30; 41 34 35; 42 23 35
36. 43 36 37; 44 24 37; 45 35 37; 46 38 39; 47 37 39; 48 26 39; 49 40 41; 50 42 43
37. 51 41 43; 52 44 45; 53 43 45; 54 35 41; 55 37 43; 56 39 45; 57 46 47; 58 48 49
38. 59 47 49; 60 50 51; 61 49 51; 62 41 47; 63 43 49; 64 45 51; 65 51 52; 66 53 54
39. 67 28 54; 68 55 56; 69 54 56; 70 57 52; 71 56 52; 72 58 59; 73 30 59; 74 60 61
40. 75 59 61; 76 62 63; 77 61 63; 78 54 59; 79 56 61; 80 52 63; 81 64 65; 82 32 65

```

41. 83 66 67; 84 65 67; 85 68 69; 86 67 69; 87 59 65; 88 61 67; 89 63 69; 90 70 71
42. 91 72 73; 92 71 73; 93 74 75; 94 73 75; 95 76 77; 96 75 77; 97 71 20; 98 73 32
43. 99 75 65; 100 77 67; 102 23 84; 103 35 85; 104 41 86; 105 47 87; 107 79 80
44. 108 80 81; 109 81 82; 111 84 79; 112 85 80; 113 86 81; 114 87 82; 116 84 85
45. 117 85 86; 118 86 87; 120 32 93; 121 65 94; 122 67 95; 123 69 96; 124 89 90
46. 125 90 91; 126 91 92; 127 93 89; 128 94 90; 129 95 91; 130 96 92; 131 93 94
47. 132 94 95; 133 95 96; 135 49 99; 136 99 98; 137 52 101; 138 101 100
48. 139 63 103; 140 103 102; 141 92 102; 142 102 100; 143 100 98; 144 98 82
49. 145 96 103; 146 103 101; 147 101 99; 148 99 87; 149 24 105; 150 105 104
50. 151 28 107; 152 107 106; 153 30 109; 154 109 108; 155 89 108; 156 108 106
51. 157 106 104; 158 104 79; 159 93 109; 160 109 107; 161 107 105; 162 105 84
52. 163 92 110; 164 100 111; 165 82 111; 166 111 110; 167 110 112; 169 89 114
53. 170 106 115; 171 79 115; 172 115 114; 173 114 116; 174 91 117; 175 81 118
54. 176 118 117; 177 117 119; 178 120 79; 179 120 80; 180 90 121; 181 80 122
55. 182 122 121; 183 121 123
56. DEFINE MATERIAL START
57. ISOTROPIC STEEL
58. E 2.05E+008
59. POISSON 0.3
60. DENSITY 76.8195
61. ALPHA 1.2E-005
62. DAMP 0.03
63. END DEFINE MATERIAL
64. CONSTANTS
65. BETA 90 MEMB 5 TO 8 12 13 17 18 22 23 25 26 29 30 34 36 41 43 46 49 50 52 -
66. 57 58 60 66 68 70 72 74 76 81 83 85 90 91 93 95 102 TO 105 111 TO 114 120 -
67. 121 TO 123 127 TO 130 135 TO 140 149 TO 154 163 164 169 170 174 180
68. MATERIAL STEEL MEMB 1 TO 100 102 TO 105 107 TO 109 111 TO 114 116 TO 118 120 -
69. 121 TO 133 135 TO 167 169 TO 183
70. MEMBER PROPERTY BRITISH
71. 1 3 4 9 TO 11 15 16 19 TO 21 39 47 53 67 69 71 97 107 TO 109 116 TO 118 124 -
72. 125 TO 126 131 TO 133 141 TO 148 155 TO 162 178 179 TABLE ST CH203X76
73. 5 TO 8 12 13 17 18 22 23 90 92 TABLE ST UB356X127X33
74. 25 26 29 30 34 36 41 43 46 49 50 52 57 58 60 66 68 70 72 74 76 81 83 85 91 -
75. 93 95 102 TO 105 111 TO 114 120 TO 123 127 TO 130 135 TO 140 149 TO 154 163 -
76. 164 169 170 174 180 TABLE ST UB533X210X82
77. 2 14 27 28 31 38 40 TABLE ST CH127X64
78. 24 32 33 35 37 42 44 45 48 51 54 TO 56 59 61 TO 65 73 75 77 TO 80 82 84 86 -
79. 87 TO 89 94 96 98 TO 100 165 TO 167 171 TO 173 175 TO 177 181 TO 182 -
80. 183 TABLE ST UB457X191X67
81. SUPPORTS
82. 1 TO 4 9 10 13 14 17 18 21 22 25 27 29 31 34 36 38 40 42 44 46 48 50 53 55 -
83. 57 58 60 62 64 66 68 70 72 74 76 PINNED
84. LOAD 1 DL
85. SELFWEIGHT Y -1
86. MEMBER LOAD
87. 3 10 15 20 UNI GY -1.2
88. 4 11 16 21 39 UNI GY -5.95
89. 24 32 33 35 37 98 UNI GY -4.75
90. 1 19 UNI GY -0.75
91. 97 UNI GY -0.75
92. 27 UNI GY -0.75
93. 92 94 96 UNI GY -0.75
94. 42 54 62 UNI GY -4.75
95. 44 55 63 UNI GY -1.05
96. 48 56 64 UNI GY -0.98

97. 67 69 71 UNI GY -0.98
98. 73 75 77 UNI GY -1.5
99. 82 84 UNI GY -5.61
100. 86 UNI GY -4.98
101. 59 61 65 80 89 UNI GY -4.4
102. 165 TO 167 171 TO 173 UNI GY -1.2
103. 175 TO 177 181 TO 183 UNI GY -2.4
104. LOAD 2 LL
105. MEMBER LOAD
106. 3 10 15 20 UNI GY -10.2
107. 4 11 16 21 39 UNI GY -13.95
108. 24 32 33 35 37 98 UNI GY -3.75
109. 1 19 UNI GY -6.38
110. 97 UNI GY -3.75
111. 27 UNI GY -3.75
112. 92 94 96 UNI GY -3.75
113. 42 54 62 UNI GY -12
114. 44 55 63 UNI GY -16.8
115. 48 56 64 UNI GY -15.6
116. 67 69 71 UNI GY -15.68
117. 73 75 77 UNI GY -24
118. 82 84 UNI GY -18.75
119. 86 UNI GY -15.6
120. 59 61 65 80 89 UNI GY -6.4
121. 165 TO 167 171 TO 173 UNI GY -0.9
122. 175 TO 177 181 TO 183 UNI GY -1.8
123. LOAD 3 WL
124. MEMBER LOAD
125. 165 TO 167 171 TO 173 UNI GY -0.9
126. 175 TO 177 181 TO 183 UNI GY -1.8
127. LOAD COMB 4 1.0DL+1.0LL
128. 1 1.0 2 1.0
129. LOAD COMB 5 1.4DL+1.6LL
130. 1 1.4 2 1.6
131. LOAD COMB 6 1.2DL+1.2LL+1.2WL
132. 1 1.2 2 1.2 3 1.2
133. LOAD COMB 7 1.2DL+1.2LL-1.2WL
134. 1 1.2 2 1.2 3 -1.2
135. PERFORM ANALYSIS

PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 118/ 176/ 38
ORIGINAL/FINAL BAND-WIDTH= 77/ 19/ 108 DOF
TOTAL PRIMARY LOAD CASES = 3, TOTAL DEGREES OF FREEDOM = 594
SIZE OF STIFFNESS MATRIX = 65 DOUBLE KILO-WORDS
REQD/AVAIL. DISK SPACE = 13.0/ 9088.9 MB, EXMEM = 218.8 MB

136. PARAMETER